

AI SIEM Platform

Complete Lab Submission Guide, Project Information, Workflow, and Technology Stack

Project summary: A full-stack Security Operations Center prototype that collects security logs, detects threats with rules, creates alerts, supports incident tracking, and provides AI-assisted analysis with offline fallback mode.

Item	Details
Application URL	http://127.0.0.1:5173
Database GUI	http://127.0.0.1:8080
Default user	admin@siem.local / changeme123
Submission purpose	Demonstrate SIEM architecture, detection rules, alert handling, incidents, and database records.

1. Project Overview

The AI SIEM Platform is a lab-ready SIEM and SOC dashboard. It simulates how security teams collect logs, detect suspicious behavior, create alerts, and investigate incidents. The platform is designed for a real-world inspired demonstration while staying safe for a local lab machine.

- Central dashboard for threat monitoring and SOC metrics.
- Log ingestion through REST APIs, collector endpoint, and syslog UDP.
- Rule-based alerting for brute force, SQL injection, credential dumping, and suspicious login chains.
- Incident management for analyst investigation and response tracking.
- AI Analysis page with offline deterministic analysis when no external API key is configured.
- PostgreSQL database GUI for showing stored users, logs, alerts, rules, incidents, and audit records.

2. Technology Stack

Layer	Technology	Purpose
Frontend	React 18, Vite, TailwindCSS, Recharts	SOC dashboard, pages, charts, forms, and analyst workflow.
Backend	Node.js, Express, Socket.IO	Authentication, APIs, ingestion, alert processing, and real-time updates.
Database	PostgreSQL 16	Stores users, logs, alerts, incidents, rules, audit logs, and AI analysis history.
Cache/Queue	Redis 7	Supports rate limiting and background processing behavior.
Security	JWT, bcrypt, Helmet, CORS, Joi	Authentication, password hashing, API hardening, and validation.
Deployment	Docker Compose	Runs frontend, backend, Postgres, Redis, and Adminer locally.
Database GUI	Adminer	Browser-based GUI for viewing database records.
AI	Anthropic SDK with offline fallback	Online Claude integration when key exists; offline lab mode otherwise.

3. System Architecture

Component	Port	Role
Frontend container	5173 -> 80	Serves the React application through Nginx.
Backend container	3001	Provides REST API, auth, ingestion, rules, alerts, and Socket.IO.
PostgreSQL container	5432	Stores persistent application data.
Redis container	6379	Supports rate limits and service coordination.
Adminer container	8080	Database GUI for lab inspection.
Syslog UDP	5514	Receives syslog-style security events.

Architecture flow: Browser -> React frontend -> Express backend API -> PostgreSQL/Redis. Security logs enter the backend, rules detect suspicious patterns, alerts are created, and analysts review results in the SOC interface.

4. Application Workflow

- 1 User logs in with the admin account.
- 2 Security events are ingested through API, collector, demo script, or syslog.
- 3 Backend normalizes and stores events in PostgreSQL.
- 4 Detection rules check each event or event pattern.
- 5 Matching rules create alerts with severity and status.
- 6 Dashboard, Logs, Alerts, Incidents, AI Analysis, Rules, and Settings pages allow analyst investigation.
- 7 Resolved alerts and incident records show response completion.

5. Main Application Modules

Module	What it shows	How to explain it
Dashboard	Threat metrics, open alerts, severity chart, source activity, asset health.	SOC overview screen.
Logs	Ingested event records with search/filtering.	Raw and normalized security evidence is stored here.
Alerts	Rule detections with acknowledge and resolve actions.	Open -> Acknowledge -> Resolve is the analyst workflow.
Incidents	Investigation records and timelines.	Analysts group related alerts/logs into a case.
AI Analysis	Offline or API-backed triage notes.	Summarizes attack type, indicators, MITRE hints, and response steps.
Rules	Detection rule configuration.	Rules convert suspicious logs into alerts.
Settings	Access details, login/logout, password change, project about.	User account and project explanation area.

6. How to Start the Project

- 1 Open PowerShell: `cd "C:\Users\bpava\OneDrive\Documents\New project\Wazuh-SIEM-Security-Lab\siem-platform"`
- 2 Start Docker Desktop and wait until the engine is running.
- 3 Run: `docker compose up -d --build`
- 4 Check status: `docker compose ps`
- 5 Open the application: `http://127.0.0.1:5173`
- 6 Login with `admin@siem.local` and password `changeme123`.

Command	Purpose
<code>docker compose up -d --build</code>	Build and start all services.
<code>docker compose ps</code>	Confirm frontend, backend, postgres, redis, and adminer are running.
<code>docker compose logs --tail 80 backend</code>	Check backend errors or login/API activity.
<code>docker compose down</code>	Stop the stack while keeping database data.
<code>docker compose down -v</code>	Reset all database data. Use only when a fresh database is required.

7. Demo Scenario for Submission

Run the attack simulation after the stack is running and login works:

Demo command: `powershell.exe -ExecutionPolicy Bypass -File .\tools\simulate-attacks.ps1 -BaseUrl http://127.0.0.1:3001`

- 1 Run the demo script and copy the Run ID shown in PowerShell.

- 2 Open Dashboard and show updated threat metrics and charts.
- 3 Open Logs and search the Run ID to show ingested events.
- 4 Open Alerts and show detections for brute force, SQL injection, suspicious login, and credential dumping.
- 5 Acknowledge an alert to show investigation started.
- 6 Resolve an alert to show the issue is handled.
- 7 Open AI Analysis and explain offline analysis mode if no real API key is configured.
- 8 Open Adminer database GUI and show logs, alerts, users, rules, incidents, and audit records.

8. Database GUI and Important Tables

Adminer field	Value
URL	http://127.0.0.1:8080
System	PostgreSQL
Server	postgres
Username	siem_user
Password	password
Database	siem_db

Table	Purpose
users	Application users, role, active status, and password rotation timestamp.
logs	Security events ingested into the SIEM.
alerts	Threat detections generated by enabled rules.
incidents	Analyst investigation and response records.
rules	Detection rules and conditions.
log_sources	Allowed log sources and API keys.
audit_log	Login, profile, password, and other important user actions.
ai_analyses	AI/offline analysis request history and output.

Database safety: Do not click Drop, Truncate, or Delete during submission unless you intentionally want to remove data.

9. Account and Password Management

The Settings page supports access management for the logged-in user. The user can update full name, update email, change password, and log out. Passwords are stored as bcrypt hashes, not plain text.

- Default account: admin@siem.local / changeme123.

- Password change requires current password, new password, and confirmation.
- The system tracks password_changed_at in the users table.
- The Settings page shows the 15-day password rotation status.
- Logout returns the user to the login screen.

10. AI Analysis Mode

The project can use a real Anthropic API key, but it is not required for the lab. If ANTHROPIC_API_KEY is empty, the application uses deterministic offline SOC analysis mode.

Mode	When used	How to explain
Offline mode	No ANTHROPIC_API_KEY in .env	Safe lab mode that still extracts attack type, indicators, MITRE hints, and response steps.
Online mode	A real Anthropic API key is configured	Calls Claude through the backend for live AI analysis.

11. Common Issues and Fixes

Issue	Fix
Docker CLI permission denied	Open Docker Desktop and run commands from the real Windows account.
localhost does not open in Codex browser	Use http://127.0.0.1:5173.
Login failed on 127.0.0.1	Backend CORS must allow both localhost and 127.0.0.1.
PowerShell blocks npm scripts	Use npm.cmd run build.
Need database GUI	Open Adminer at http://127.0.0.1:8080.
Need fresh database	Use docker compose down -v, then rebuild and seed. This deletes existing data.

12. Viva Explanation Script

Short answer: This is an AI-assisted SIEM platform. It collects logs, stores them in PostgreSQL, applies detection rules, creates alerts, supports incidents, and provides SOC dashboard views for investigation.

- Acknowledge means the analyst has started investigating an alert.
- Resolve means the alert has been handled and closed.
- Threats Blocked is a dashboard metric showing detected/handled malicious activity.
- Offline AI mode is used because a real paid API key is not required for the lab.
- Adminer proves the backend data is actually stored in PostgreSQL.

13. Final Submission Checklist

- Docker Desktop is running.
- `docker compose ps` shows all containers running.
- Frontend opens at `http://127.0.0.1:5173`.
- Login works with `admin@siem.local`.
- Demo attack simulation has been run.
- Logs and alerts are visible in the UI.
- Adminer opens and shows PostgreSQL tables.
- Settings shows access, login/logout, password change, and about project.